

The Mathematical Model: Differential Equations

$$\frac{dC_i}{dt} = \alpha_i C_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j C_j - k_i C_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

$$E_i = \lambda_i C_i$$

$$i \in \{1, 2, \dots, N_{Regions}\}$$

$$P_i(t), N_i(t) > 0$$

$$E_i, C_i > 0$$

$$\alpha_i, k_i, \lambda_i > 0$$

Finding $P_i(t)$ and $N_i(t)$:

We will find $N_i(t)$ via linear regression using the data points on number of citizens in region i from 2021 to 2026 and $P_i(t)$ via linear interpolation using the police population data points from 2021 to 2026.

Finding α_i, k_i, α_j :

Possible options 1), 2) and 3):

1) General Least squares solution:

Substitute $C_i = \frac{E_i}{\lambda_i}$ to $\frac{dC_i}{dt}$:

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j=1, j \neq i}^{N_R} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

$B(i)$ is the set of j -th regions, $j \in B(i)$, which i -th region borders j -th region. Take $\alpha_j = 0$ when $j \notin B(i)$. (modeling assumption). So the equation becomes:

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j \in B(i)} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

We know the data needs to satisfy: (for any month but month 1 month 2 as an example)

$$\int_{m_1}^{m_2} \frac{dE_i}{dt} dt = E_i(m_2) - E_i(m_1)$$

Simultaneous Integral Equations:

$$\int_{m_1}^{m_2} \frac{dE_i}{dt} dt = E_i(m_2) - E_i(m_1)$$

$$\int_{m_2}^{m_3} \frac{dE_i}{dt} dt = E_i(m_3) - E_i(m_2)$$

...

$$\int_{m_{K-1}}^{m_K} \frac{dE_i}{dt} dt = E_i(m_K) - E_i(m_{K-1})$$

Which can be expressed using matrices and vectors:

$$\begin{bmatrix} E_i(m_2) - E_i(m_1) \\ E_i(m_3) - E_i(m_2) \\ \dots \\ E_i(m_K) - E_i(m_{K-1}) \end{bmatrix} = \begin{bmatrix} \int_{m_1}^{m_2} E_i dt & \int_{m_1}^{m_2} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_1}^{m_2} E_j dt & \dots & \int_{m_1}^{m_2} E_j dt \\ \int_{m_2}^{m_3} E_i dt & \int_{m_2}^{m_3} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_2}^{m_3} E_j dt & \dots & \int_{m_2}^{m_3} E_j dt \\ \dots & \dots & \dots & \dots & \dots \\ \int_{m_{K-1}}^{m_K} E_i dt & \int_{m_{K-1}}^{m_K} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_{K-1}}^{m_K} E_j dt & \dots & \int_{m_{K-1}}^{m_K} E_j dt \end{bmatrix} \begin{bmatrix} \alpha_i \\ k_i \\ \alpha_j \dots \\ \alpha_{N_R} \end{bmatrix}$$

Which is a linear least squares problem:

$$\overline{b}_i = A_i \overline{x}_i$$

Which has the solution formula: (where superscript -1 means matrix inverse and superscript T means transpose of the matrix)

$$\overline{x}_i = (A_i^T A_i)^{-1} A_i^T \overline{b}_i$$

Where:

$$\overline{b}_i = \begin{bmatrix} E_i(m_2) - E_i(m_1) \\ E_i(m_3) - E_i(m_2) \\ \dots \\ E_i(m_K) - E_i(m_{K-1}) \end{bmatrix}$$

$(K - 1) \times 1$ vector

$$A_i = \begin{bmatrix} \int_{m_1}^{m_2} E_i dt & \int_{m_1}^{m_2} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_1}^{m_2} E_j dt & \dots & \int_{m_1}^{m_2} E_j dt \\ \int_{m_2}^{m_3} E_i dt & \int_{m_2}^{m_3} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_2}^{m_3} E_j dt & \dots & \int_{m_2}^{m_3} E_j dt \\ \dots & \dots & \dots & \dots & \dots \\ \int_{m_{K-1}}^{m_K} E_i dt & \int_{m_{K-1}}^{m_K} E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right) dt & \int_{m_{K-1}}^{m_K} E_j dt & \dots & \int_{m_{K-1}}^{m_K} E_j dt \end{bmatrix}$$

$(K - 1) \times (N_R + 1)$ matrix

$$\overline{x}_i = \begin{bmatrix} \alpha_i \\ k_i \\ \alpha_j \dots \\ \alpha_{N_R} \end{bmatrix}$$

$(N_R + 1) \times 1$ vector

The integrals in the matrices A_i can be computed via Trapezoid rule:

$$\int_a^b f(x) dx = \frac{f(a) + f(b)}{2} (b - a), \quad b > a$$

(this is a respected numerical approximation method, we kind of assume and also know (criminal population is not prone to nonlinear changes in a month) that $f(x)$ does not behave too extreme)

E_i does not need to be interpolated but $P_i(t)$ and $N_i(t)$ does (or linear regression). My suggestion is linear interpolation for $P_i(t)$ and linear regression for $N_i(t)$

2) Incorporating Deprivation Index

Same steps until here:

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j \in B(i)} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

Take $\alpha_i = \text{Deprivation Index}_i$

Take $\alpha_j = \text{Deprivation Index}_j$

Doing a slight constant shift to the equation by scaling with $\frac{1}{|B(i)|}$: (Because we assume each bordering region has equivalent border effect but of course different deprivation data)

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j \in B(i)} \frac{1}{|B(i)|} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

$|B(i)|$: Number of elements in the set $B(i)$.

Assumption: Deprivation Indexes do not change much in 5-10 years. It is a long term result thus it is a constant, it does not change over time. No need to interpolate it at all, data from one year is enough.

Now we force k_i out using Least Squares method like in option 1):

$$\int_{m_1}^{m_2} \frac{dE_i}{dt} dt = E_i(m_2) - E_i(m_1),$$

Simultaneous Integral Equations:

$$\int_{m_1}^{m_2} \frac{dE_i}{dt} dt = E_i(m_2) - E_i(m_1)$$

$$\int_{m_2}^{m_3} \frac{dE_i}{dt} dt = E_i(m_3) - E_i(m_2)$$

...

$$\int_{m_{K-1}}^{m_K} \frac{dE_i}{dt} dt = E_i(m_K) - E_i(m_{K-1})$$

$$\bar{b}_i = A_i \bar{x}_i$$

$$\bar{x}_i = (A_i^T A_i)^{-1} A_i^T \bar{b}_i$$

But this time $\alpha_i, \frac{1}{|B(i)|} \alpha_j$ are known we solve for k_i

3) Similar Thinking to 2) but instead we incorporated the data of average number of crimes that end up with "Waiting for Court Decision" divided by average number of crimes and force α_i, α_j out

$$\frac{dE_i}{dt} = \alpha_i E_i + \sum_{j \in B(i)} \alpha_j E_j - k_i E_i \left(\frac{P_i(t)}{N_i(t) + P_i(t)} \right)$$

4) Your ideas